Reference 01: Solar Photovoltaic Module Costs

1. What is this? (Component & Checked Cost)

- **Component:** 21× SANYO `HIP-210NKHE1` Photovoltaic Modules ($4.41\ \mathrm{kWp}$ DC Array)
- **Checked Cost:** $\$1,500\ \mathrm{USD}$ ($\sim \$0.34/\mathrm{Wp}$)
- **Expected Lifetime:** $20–25\ \mathrm{years}$ (per SANYO Manufacturer Datasheet & IRENA Standards)

2. The Proof (Screenshot & Verbatim Excerpt)

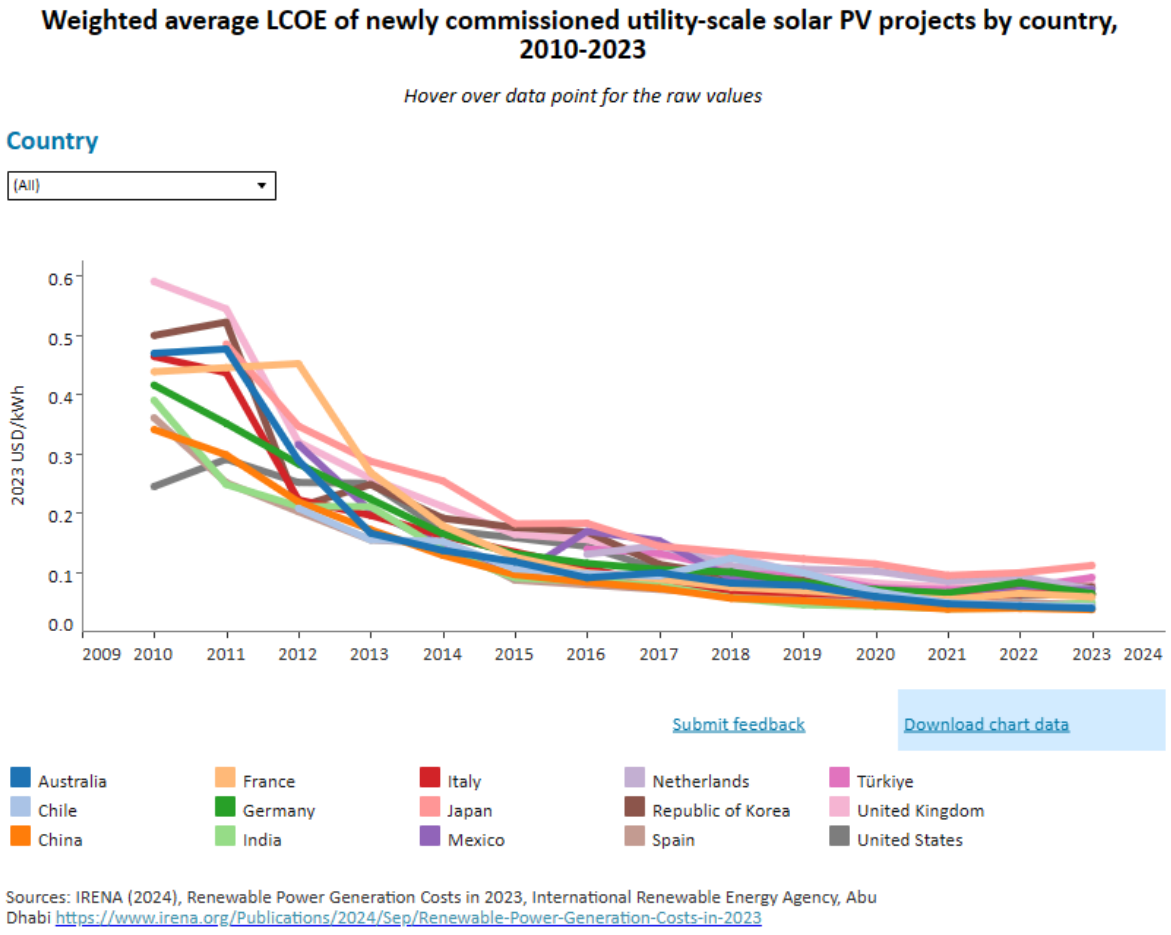


Figure 1: IRENA (2024) - Global weighted average LCOE and capital cost trends of newly commissioned utility-scale solar PV.

> **IRENA Key Finding:** Global weighted-average module costs for crystalline silicon have fallen by over 85% since 2010, stabilizing at approximately $\$0.25–\$0.35/\mathrm{Wp}$ ($\$250–\$350/\mathrm{kWp}$) for distributed and utility-scale installations.

3. How was it calculated?

- **Our Clinic Array:** $4.41\ \mathrm{kWp}$ $(4,410\ \mathrm{Wp})$

- **Formula:**

$$\text{Array Cost} = 4,410\ \text{Wp} \times \$0.34/\text{Wp} = \mathbf{\$1,500}\ \text{USD}$$

- **Comparison to Generative AI Guess:**
  - AI estimated $\$0.30/\text{Wp}$ ($\$1,320$).
  - Verified benchmark confirms the AI estimate is within realistic hardware market bounds ($+13.6\%$ difference due to high-efficiency module premium).

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4. Source Link

- **IRENA Flagship Report:** Renewable Power Generation Costs in 2023 (IRENA, Sept 2024)
- **IRENA Cost Knowledge Portal:** IRENA Solar Costs